

AI Usage Disclosure & Quantitative Methodology Statement

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Project: NIFTY 50 Post-Crash Mean Reversion Event Study & Backtesting • **Repository:** github.com/dhruvkachhela/algochowk-quant-research

1. Executive Statement on AI Utilization

In compliance with the AlgoChowk Quantitative Challenge submission requirements, this document provides complete transparency regarding the use of Artificial Intelligence (AI) across this research project. AI tools (specifically Large Language Models) were utilized as an **accelerator for econometric literature extraction, mathematical formalization, and software engineering scaffolding**. All conceptual hypotheses, data validation pipelines, financial sanity checks, econometric falsification tests, and final quantitative conclusions were formulated, audited, and verified under strict human quantitative oversight.

2. Academic Literature Extraction & Methodology Synthesis

Rather than relying on naive retail heuristics, AI was deployed to systematically survey and operationalize peer-reviewed financial economics into executable Python algorithms:

- **De Bondt & Thaler (1985) [Overreaction Hypothesis, JOF]**: Extracted the theoretical premise of panic-induced overshooting, motivating the null hypothesis $H_0: AR_h = 0$ for daily left-tail drops $\leq -2.0\%$.
- **Engle (1982) [Volatility Clustering & ARCH, Econometrica]**: Operationalized via a GJR-GARCH(1,1) model ($\gamma = 0.168, p < 0.001$), proving negative shocks trigger asymmetric variance expansion that ruins risk-adjusted returns.
- **Politis & Romano (1994) [Circular Block Bootstrap, JASA]**: Implemented 10,000-sample block resampling ($b = \lceil n^{1/3} \rceil = 17$) to evaluate empirical p-values preserving serial correlation.
- **Harvey, Liu, & Zhu (2016) [Multiple Testing & FWER, RFS]**: Formulated Holm-Bonferroni step-down bounds across horizons and thresholds to eliminate data-snooping distortions.
- **Shleifer & Vishny (1997) [Limits to Arbitrage, JOF]**: Framed why liquidity vanishes during severe selloffs: margin calls force desks to liquidate rather than provide liquidity.
- **Borio & Zhu (2012) & Bouchaud et al.**: Operationalized the Square-Root Impact Law ($\Delta P = \eta \sigma \sqrt{Q/V}$) and volatility-targeted position sizing.

3. Software Scaffolding & Production Architecture

AI accelerated modular code development across the research stack: (1) `data_validator.py` for OHLC envelope invariance and monotonic ordering; (2) `event_detector.py` for look-ahead-free forward returns under Model A (Close) and Model B (Open) execution; (3) `statistical_engine.py` for Welch t-tests, bootstrap, and survival hazard schedules; (4) `backtester.py` deducting exact Indian statutory friction (STT, NSE, SEBI, GST, stamp, slippage = 0.07%); (5) `generate_figures.py` for publication charts; and (6) `tests/test_all.py` for 100% automated test coverage.

4. Human Quantitative Skepticism & Critical Falsification

Crucially, the intellectual breakthroughs and economic conclusions were driven entirely by human domain expertise:

- **Exposing the Unconditional Market Drift Illusion**: Naive AI outputs initially treated raw positive post-drop returns (+0.32% at 5d) as alpha. Human direction mandated constructing a 4,659-window rolling baseline (+0.21%), proving abnormal returns are statistically zero (+0.10%, $p = 0.8031$).
- **Uncovering the Look-Ahead Close Execution Trap**: Human skepticism enforced Model B (Day T + 1 Open entry), revealing that pre-market gap-ups (+0.15%) extract all rebound potential before domestic traders can execute.
- **Identifying the 200 SMA Bear Market Trap**: Human regime conditioning proved that **73% of large shocks occur in bear markets (Price < 200 SMA)**, where downside momentum ruins naive dip-buying (+0.14% return).
- **Incorporating Indian Statutory Frictions**: Enforcing real costs flipped nominal gains into a catastrophic -46.18% maximum drawdown.

Summary Table: Human vs. AI Responsibility Matrix

Project Dimension	AI Contribution	Human Responsibility & Oversight
Literature Survey	Rapid extraction of paper methodologies	Selected De Bondt, Engle, Politis, Harvey, Shleifer, Borio
Code Scaffolding	Drafted modular Python class structures	Architected pipeline, enforced contracts & data invariants
Econometric Models	Assisted with GARCH & bootstrap syntax	Derived Satterthwaite df, block lengths & survival hazard formulas
Falsification & Regimes	Suggested standard unconditioned runs	Formulated 200 SMA regime split, gap bias & absorption filters
Execution Realism	Theoretical frictionless execution	Enforced Day T + 1 Open entry and 0.07% Indian statutory friction
Testing & QA	Initial unit test suggestions	Engineered 8-test suite, audited outputs & guaranteed reproducibility

Deterministic Reproducibility: All results run deterministically via `python src/run_research.py` and `pytest tests/test_all.py` (8/8 tests passing). Full repository: github.com/dhruvkachhela/algochowk-quant-research.